MATH 2102 Linear Algebra II

Chapter 5
More About Eigenvalues and Eigenvectors

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Chapter 5: More About Eigenvalues and Eigenvectors

A1. Setting the scene

In this chapter, we continue to assume that V and W are finite-dimensional inner product spaces over $\mathbb{F}$ (and work with the standard inner product in case the inner product space is $\mathbb{F}^n$), unless the content suggests otherwise.

A2. Orthogonal diagonalisation

One of the main goals in studying eigenvalues and eigenvectors for an operator $T \in \mathcal{L}(V)$ is to find a basis of V containing eigenvectors of T. If such basis exists, T is said to be diagonalisable.

If V is an inner product space, it would be even nicer if we could find an orthonormal basis of V containing eigenvectors of T. If such basis exists, T is said to be unitarily diagonalisable, or in case $\mathbb{F} = \mathbb{R}$, orthogonally diagonalisable.

A3. Eigenvalues and eigenvectors of self-adjoint and normal operators

$T = T^*$

(a) Let $T \in \mathcal{L}(V)$ be self-adjoint. Then every eigenvalue of T is real.

Proof:

Suppose $T(\mathbf{v}) = \lambda \mathbf{v}$ where $\mathbf{v} \neq \mathbf{0}$. Consider $\langle T(\mathbf{v}), \mathbf{v} \rangle$.

$$\langle Tv, v \rangle = \langle \lambda v, v \rangle = \lambda \|v\|$$
$$= \langle v, T^* v \rangle$$
$$= \langle v, Tv \rangle \quad \rightarrow \text{ i.e. } \lambda = \bar{\lambda} \rightarrow \lambda \in \mathbb{R}.$$
$$= \bar{\lambda} \|v\|$$

A3. Eigenvalues and eigenvectors of self-adjoint and normal operators

$TT^* = T^*T$,

(b) Let $T \in \mathcal{L}(V)$ be normal. Then

$$\lambda \text{ is an eigenvalue of } T \quad \text{if and only if} \quad \overline{\lambda} \text{ is an eigenvalue of } T^*.$$

Proof:

For a non-zero vector $\mathbf{v}$, we want to show that $T(\mathbf{v}) = \lambda\mathbf{v} \iff T^*(\mathbf{v}) = \overline{\lambda}\mathbf{v}$.

Note that for any $k \in \mathbb{F}$, $T - kI$ is also normal since

$\langle (T - \lambda I)v, v \rangle = 0 = \langle v, (T^* - \bar{\lambda} I)v \rangle$

A3. Eigenvalues and eigenvectors of self-adjoint and normal operators

(c) Let $T \in \mathcal{L}(V)$ be normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

Proof:

Suppose $T(\mathbf{v}_1) = \lambda_1 \mathbf{v}_1$ and $T(\mathbf{v}_2) = \lambda_2 \mathbf{v}_2$ where $\lambda_1 \neq \lambda_2$ and $\mathbf{v}_1, \mathbf{v}_2 \neq \mathbf{0}$. Now

$$\langle T(\mathbf{v}_1), \mathbf{v}_2 \rangle = \langle \mathbf{v}_1, T^*(\mathbf{v}_2) \rangle$$

$$= \langle \lambda v_1, v_2 \rangle \qquad = \langle v_1, \bar{\lambda}_2 v_2 \rangle$$

$$= \lambda_1 \langle v_1, v_2 \rangle \qquad = \lambda_2 \langle v_1, v_2 \rangle$$

$$\langle v_1, v_2 \rangle = 0$$

A4. Spectral theorem on $\mathbb{R}$

Lemma

Let $T \in \mathcal{L}(V)$ be self-adjoint. Then there exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is upper triangular.

(con) $\therefore m_T \cdot (T^2 - bT + cI)^{-1}$ has less degree but still equal to 0.

Proof:

We only have to consider the case $\mathbb{F} = \mathbb{R}$. If the result is not true, then m_T has a quadratic factor $t^2 - bt + c$ where $b^2 < 4c$. This would lead to a contradiction since

Lemma: T is self-adjoint $\longleftrightarrow$ $\forall b^2 < 4c$, $T^2 + bT + cI$ is invertible.

$$\begin{aligned} \forall v \in V, \langle (T^2+bT+c)v, v\rangle &= \langle Tv, Tv\rangle + b\langle Tv, v\rangle + c\langle v, v\rangle \\ &\geq \|Tv\|^2 - |b|\langle Tv, v\rangle + c\|v\|^2 \\ &\geq \|Tv\|^2 - |b| \cdot \|Tv\|\|v\| + c\|v\|^2 \\ &= \left(\|Tv\| - \frac{|b|\|v\|}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)\|v\|^2 > 0. \end{aligned}$$

i.e. $\nexists v \neq 0$ s.t. $T^2 + bT + cI = 0 \to \sim$ is injective $\to \sim$.

A4. Spectral theorem on $\mathbb{R}$

Spectral theorem on $\mathbb{R}$

Suppose $T \in \mathcal{L}(V)$ and $\mathbb{F} = \mathbb{R}$. Then T is orthogonally diagonalisable if and only if T is self-adjoint.

Proof:

($\Rightarrow$) Easy.

Gram-Schmidt process.

($\Leftarrow$) Take a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}$ is upper triangular. Turn $\mathcal{B}$ into an orthonormal basis $\mathcal{E}$. Then $[T]_{\mathcal{E}}$ is upper triangular, and hence diagonal, since

and $[T]_\mathcal{E} = [T^*]_\mathcal{E} = [T]_\mathcal{E}^t$

A5. Spectral theorem on $\mathbb{C}$

Spectral theorem on $\mathbb{C}$

Suppose $T \in \mathcal{L}(V)$ and $\mathbb{F} = \mathbb{C}$. Then T is unitarily diagonalisable if and only if T is normal.

Proof:

($\Rightarrow$) Easy.

($\Leftarrow$) Take a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}$ is upper triangular. Turn $\mathcal{B}$ into an orthonormal basis $\mathcal{E}$. Then $[T]_{\mathcal{E}}$ is upper triangular, and hence diagonal, since

$$[T]_{\mathcal{E}} = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots \\ 0 & a_{2,2} & \cdots \\ \cdots & & \cdots \end{bmatrix}$$

$$\|Te_1\| = |a_{1,1}|^2$$

$$\|T^* e_1\| = |a_{1,1}|^2 + |a_{1,2}|^2 \cdots = |Te_1|$$

$\to 0 \quad \to 0$

A6. Some numerical examples

Consider $T : \mathbb{R}^3 \to \mathbb{R}^3$ with standard matrix $\begin{bmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{bmatrix}$. We have

$$\begin{aligned} T(1,-1,0) &= 27(1,-1,0) \\ T(1,1,1) &= 9(1,1,1) \\ T(1,1,-2) &= -15(1,1,-2) \end{aligned}$$

A6. Some numerical examples

Consider $T : \mathbb{C}^2 \to \mathbb{C}^2$ with standard matrix $\begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix}$. We have

$$\begin{aligned} T(i,1) &= (2+3i)(i,1) \\ T(-i,1) &= (2-3i)(-i,1) \end{aligned}$$

Chapter 5: More About Eigenvalues and Eigenvectors

B1. An intuitive overview

We can intuitively associate operators on V to complex numbers via their eigenvalues. Different operations and types of operators are thus associated with the corresponding operations and types of complex numbers as follows:

$$\begin{aligned}
\text{Adjoint} &\quad\longleftrightarrow\quad \text{Complex conjugation}\\
\text{Self-adjoint operators} &\quad\longleftrightarrow\quad \text{Real numbers}\\
\text{Unitary operators} &\quad\longleftrightarrow\quad \text{Complex numbers with modulus 1}\\
\text{Positive operators} &\quad\longleftrightarrow\quad \text{Non-negative real numbers}
\end{aligned}$$

B2. Eigenvalues of unitary operators

Let $T \in \mathcal{L}(V)$ be unitary. Then every eigenvalue of T has modulus 1.

Proof:

Suppose $T(\mathbf{v}) = \lambda\mathbf{v}$ where $\mathbf{v} \neq \mathbf{0}$. Since a unitary operator is an isometry, we have

$$|\langle Tv, v\rangle| = |\langle \lambda v, v\rangle| = |\lambda| \, \|v\| = \|v\|$$

$$|\lambda| = 1$$

B2. Eigenvalues of unitary operators

Since a unitary operator $T \in \mathcal{L}(V)$ is normal, it follows from the spectral theorem that it is unitarily diagonalisable. Together with the previous result, we know that there exists an orthonormal basis $\mathcal{E}$ of V such that $[T]_{\mathcal{E}}$ is a diagonal matrix whose diagonal entries has modulus 1. Is the converse true?

B3. Eigenvalues of positive operators

Let $T \in \mathcal{L}(V)$ be positive. Then every eigenvalue of T is non-negative.

Proof:

Suppose $T(\mathbf{v}) = \lambda \mathbf{v}$ where $\mathbf{v} \neq \mathbf{0}$. Since T is positive, we have

$$\langle T(\mathbf{v}), \mathbf{v} \rangle \geq 0$$

$\lambda \| v \| \geq 0$

$\lambda \geq 0$

B3. Eigenvalues of positive operators

Since a positive operator $T \in \mathcal{L}(V)$ is self-adjoint (hence normal), it follows from the spectral theorem that it is unitarily diagonalisable. Together with the previous result, we know that there exists an orthonormal basis $\mathcal{E}$ of V such that $[T]_{\mathcal{E}}$ is a diagonal matrix whose diagonal entries are positive. Is the converse true?

B4. Uniqueness of square root

Let $T \in \mathcal{L}(V)$ be positive. Then T has a unique positive square root.

Proof:

We have already proved the existence, so let R be a positive square root of T. It suffices to show that whenever $\mathbf{v}$ is a λ-eigenvector of T, we have $R(\mathbf{v}) = \sqrt{\lambda}\mathbf{v}$.

Let $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n\}$ be an orthonormal basis of V consisting of eigenvectors of R, corresponding to eigenvalues $\sqrt{\lambda_1}$, $\sqrt{\lambda_2}$, ..., $\sqrt{\lambda_n}$. Write

$$\mathbf{v} = c_1\mathbf{e}_1 + c_2\mathbf{e}_2 + \cdots + c_n\mathbf{e}_n.$$

Chapter 5: More About Eigenvalues and Eigenvectors

C1. Definition

Let $T \in \mathcal{L}(V, W)$. The singular values of T are the non-negative square roots of the eigenvalues of $T^* T$.

Remarks: ①

and self-adjoint

- $T^* T$ is a positive operator so the above definition makes sense.
- As a convention, the singular values of T are usually written in descending order.
- The multiplicity of a singular value of T is taken to be the dimension of the corresponding eigenspace of $T^* T$.

we have:

①

7.64 T^*T 的性质

设 $T \in \mathcal{L}(V, W)$，那么

(a) T^*T 是 V 上的正算子；

(b) $\operatorname{null} T^*T = \operatorname{null} T$；

(c) $\operatorname{range} T^*T = \operatorname{range} T^*$；

(d) $\dim \operatorname{range} T = \dim \operatorname{range} T^* = \dim \operatorname{range} T^*T$.

② num of T's singular value $= \dim T^*T$

num of T's non-zero ~ $= \dim \operatorname{range} T$

④ T is normal $\to s_i = 1$

$T^*T = I$

C2. Examples

$T^*T(a,b,c,d) = (9a, 4b, 0, 9d)$

Find the singular values of the following:

$M(T^*T) = \begin{bmatrix} 9 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}$

(a) $T : \mathbb{F}^4 \to \mathbb{F}^4$ defined by

$\underline{3, 3, 2, 0}$

$$T(a,b,c,d) = (0, 3a, 2b, -3d)$$

$M(T) = \begin{bmatrix} 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$ $\quad M(T^*) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$

C2. Examples

Find the singular values of the following:

(b) $T : \mathbb{F}^4 \to \mathbb{F}^3$ defined by

$T^*T(a,b,c,d) = (a+b, a+b, 0, 25d)$

$$T(a, b, c, d) = (-5d, 0, a + b)$$

$5, \sqrt{2}, 0\ 0.$

$\begin{bmatrix} 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ -5 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 25 \end{bmatrix}$

$(a+b)^2 = c^2 + (d$

$\frac{a+b}{a} = \frac{a+b}{b}$

$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ 25 $\quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ 0 $\quad \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ 2

C3. Singular value decomposition (SVD)

Suppose $T \in \mathcal{L}(V, W)$ and s_1, s_2, ..., s_r are all the positive singular values of T.

SVD of an operator

There exist orthonormal sets $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_r\} \subseteq V$ and $\{\mathbf{f}_1, \mathbf{f}_2, ..., \mathbf{f}_r\} \subseteq W$ such that

$$T(\mathbf{v}) = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_r \langle \mathbf{v}, \mathbf{e}_r \rangle \mathbf{f}_r.$$

1. $(T^*T)^* = (T)^*(T^*)^* = T^*T \to T^*T$ self adjoint.

By spectral theorem, $\exists$ ob is T^*T's eigenvalues.

i.e. $\forall i$, $T^*Te_i = s_i^2 e_i$ Let $f_i = \frac{Te_i}{s_i}$, then,

$T(v) = T(\Sigma \langle v, e_i \rangle e_i) = \sim$.

and $\forall i, j$ $f_i \cdot f_j = \frac{1}{s_i s_j} \langle Te_i, Te_j \rangle$

C3. Singular value decomposition (SVD)

To prove the result on the previous page, note that by the spectral theorem there exists an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ of V such that

$$(T^*T)(\mathbf{e}_k) = s_k^2\mathbf{e}_k \quad \text{for every } k \in \{1, 2, \ldots, n\}.$$

Take $\mathbf{f}_k = \dfrac{1}{s_k}T(\mathbf{e}_k)$ for $k \in \{1, 2, \ldots, r\}$. We check that

- the set $\{\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_r\}$ is orthonormal;
- $(T^*T)(\mathbf{e}_k) = \mathbf{0}$ if $k > r$; and
- $T(\mathbf{v}) = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_r \langle \mathbf{v}, \mathbf{e}_r \rangle \mathbf{f}_r$.

C3. Singular value decomposition (SVD)

Consider the linear map $T : \mathbb{F}^4 \to \mathbb{F}^3$ defined by

$$T(a, b, c, d) = (-5d, 0, a + b).$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 25 \end{bmatrix}$$

Can you find the SVD of T?

$$\begin{array}{cccc} 5 & \sqrt{2} & 0 & 0 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} & \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix} & \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \end{array}$$

$$\to \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \quad \to \begin{pmatrix} 0 \\ 0 \\ \sqrt{2} \end{pmatrix} \quad \to \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \to \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$T(v) = 5\left\langle v, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + \sqrt{2}\left\langle v, \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle \begin{pmatrix} 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

C4. SVD for adjoint and pseudoinverse

Suppose $T \in \mathcal{L}(V, W)$ and the SVD of T is

$$T(\mathbf{v}) = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_{\cancel{m}\,r} \langle \mathbf{v}, \mathbf{e}_r \rangle \mathbf{f}_r .$$

Can you work out the SVD of T^*?

$$T^* T(v) = \sum s_r \langle v, e_r \rangle \cancel{T^*} f_r = \sum s_r^2 \langle v_r, e_r \rangle e_r$$

$$T^* T e_r = s_r^2 e_r$$

$$\frac{T e_r}{s_r} = f_r$$

$$T^* (w) = \sum s_r \langle w, f_r \rangle e_r$$

$$\frac{\cancel{T^*} T e_r}{\cancel{s_r}} \quad s_r e_r$$

$$T T^* f_r = s_r T e_r = s_r^2 f_r$$

$$\frac{T^* f_r}{s_r} = \frac{T^* T e_r}{s_r^2} = e$$

C4. SVD for adjoint and pseudoinverse

Suppose $T \in \mathcal{L}(V, W)$ and the SVD of T is

$$T(\mathbf{v}) = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_m \langle \mathbf{v}, \mathbf{e}_r \rangle \mathbf{f}_r.$$

Can you work out the SVD of $T^\dagger$?

$T^\dagger f_r = \frac{T^\dagger T e_r}{s_r} = \frac{e_r}{s_r}$

$(T^\dagger)^* T^\dagger$

M

SVD of matrix.

$A \to p \times n$, rank $A = m \geq 1$.

$\exists$ orthonormal matrix $B \to p \times m$, $C \to n \times m$.

all positive element diagonal D

$A = BDC^*$

$B: [f_1, \cdots f_m]$

$D: \begin{bmatrix} s_1 & & 0 \\ & \ddots & \\ 0 & & s_m \end{bmatrix}$ $C: [e_1, \cdots e_m]$

C5. Norm of a linear map

Given $T \in \mathcal{L}(V, W)$, define the norm of T, denoted $\|T\|$, by

$$\|T\| = \max\{\|T(\mathbf{v})\| : \mathbf{v} \in V \text{ and } \|\mathbf{v}\| = 1\}.$$

It is not hard to see that

largest

$\|Tv\|^2 = \sum s_i^2 |\langle v, e_i\rangle|^2 \|e_i\|^2 \le s_1^2 \left(\sum |\langle v, e_i\rangle|^2\right) = s_1^2 \|v\|^2$

- $\|T\|$ is equal to the largest singular value of T;
- $\|T\| \geq 0$, equality if and only if $T = 0$;
- $\|\lambda T\| = |\lambda| \|T\|$;
- $\|S + T\| \leq \|S\| + \|T\|$;
- $\|T^*\| = \|T\|$;
- a unitary operator has norm 1.

$\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, T^*Tv\rangle$

$\le \|v\| \|T^*Tv\|$

$\le \|v\| \|T^*\| \|Tv\|$

$\to \|Tv\| \le \|T^*\| \|v\|$

$\|T\| = \|Te\| \le \|T^*\| \|e\| = \|T^*\|$

verse vice.

C6. Polar decomposition

$z = r \cdot e^{i\theta}$

Let $T \in \mathcal{L}(V)$. A polar decomposition of T is a factorisation of T in the form

$$T = S\sqrt{T^*T}$$

where S is a unitary operator on V.

$Sv = \sum \langle v, e_i \rangle f_i$

$T^*Tv = \sum s_r^2 \langle v, e_r \rangle e_r$

$\therefore \; S\sqrt{T^*T}\,v = S \sum s_r \langle v, e_r \rangle e_r = Tv.$

C6. Polar decomposition

We shall construct a polar decomposition for an arbitrary T. Suppose T has SVD

$$T(\mathbf{v}) = s_1 \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + s_r \langle \mathbf{v}, \mathbf{e}_r \rangle \mathbf{f}_r.$$

Separately extend $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_r\}$ and $\{\mathbf{f}_1, \mathbf{f}_2, ..., \mathbf{f}_r\}$ to bases of V. Define

$$S(\mathbf{v}) = \langle \mathbf{v}, \mathbf{e}_1 \rangle \mathbf{f}_1 + \cdots + \langle \mathbf{v}, \mathbf{e}_n \rangle \mathbf{f}_n.$$

Chapter 5: More About Eigenvalues and Eigenvectors

D1. Setting the scene again

For the rest of this chapter we assume $T \in \mathcal{L}(V)$ and $\dim V = n > 0$, unless otherwise specified.

Moreover, the results in the rest of this chapter do not depend on an inner product. Hence we can simply regard V to be a finite-dimensional vector space over $\mathbb{F}$.

D2. Some preparatory work

$\forall v \in \ker T^m$, $T^{m+1} v = T(T^m v) = 0$.

(a) $\ker T^m \subseteq \ker T^{m+1}$ for all non-negative integers m.

(b) $\ker T^m = \ker T^{m+k}$ for all $k \in \mathbb{N}$ whenever $\ker T^m = \ker T^{m+1}$ for some non-negative integer m.

D2. Some preparatory work

(c) $\ker T^n = \ker T^{n+k}$ for all $k \in \mathbb{N}$.

(d) $V = \ker T^n \oplus \operatorname{im} T^n$.

D3. Definition

Let λ be an eigenvalue of T. We say that a non-zero vector $\mathbf{v}$ is a generalised eigenvector of T corresponding to λ if $(T - \lambda I)^k(\mathbf{v}) = \mathbf{0}$ for some positive integer k.

Note that

A non-zero vector $\mathbf{v}$ is a generalised eigenvector of T if and only if

$$(T - \lambda I)^n(\mathbf{v}) = \mathbf{0}.$$

D3. Definition

As an example, consider the operator $T \in \mathcal{L}(\mathbb{F}^3)$ defined by

$$T(x, y, z) = (4y, 0, 5z).$$

$$\mathcal{M}(T) = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix} \qquad \mathcal{M}(T - \lambda I) = \begin{bmatrix} -\lambda & 0 & 0 \\ 4 & -\lambda & 0 \\ 0 & 0 & 5-\lambda \end{bmatrix}$$

$$0 \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad 5 \to \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{bmatrix} \lambda^2 & 0 & 0 \\ -8\lambda & \lambda^2 & 0 \\ 0 & 0 & (5-\lambda)^2 \end{bmatrix} \qquad \begin{bmatrix} -\lambda^3 & 0 & 0 \\ 12\lambda^2 & -\lambda^3 & 0 \\ 0 & 0 & (5-\lambda)^3 \end{bmatrix}$$

$$\begin{matrix} (-\lambda)^n & 0 & 0 \\ 4(-\lambda)^{n-1} & (-\lambda)^n & 0 \\ 0 & 0 & (5-\lambda)^n \end{matrix} \qquad 0 - 100$$

D4. Properties

(a) If $\mathbb{F} = \mathbb{C}$, there exists a basis of V consisting of generalised eigenvectors of T.

Proof:

We prove by induction on n. The result clearly holds for $n = 1$. Suppose $n > 1$ and the result holds for smaller n. Take an eigenvalue λ of T. Then

$$V = \ker(T - \lambda I)^n \oplus \operatorname{im}(T - \lambda I)^n.$$

D4. Properties

(b) If $\mathbf{v}$ is a generalised eigenvector of T, then $\mathbf{v}$ corresponds to a unique eigenvalue.

Proof:

Suppose $\mathbf{v}$ corresponds to both eigenvalues λ and λ'. Take the smallest m for which $(T - \lambda' I)^m(\mathbf{v}) = \mathbf{0}$.

D4. Properties

(c) Generalised eigenvectors of T corresponding to distinct eigenvalues are linearly independent.

Proof:

Take a smallest counterexample, say, T has generalised eigenvectors $\mathbf{v}_1$, $\mathbf{v}_2$, ..., $\mathbf{v}_m$ which are linearly dependent, and they correspond to distinct eigenvalues λ_1, λ_2, ..., λ_m of T. Then there exist scalars c_1, c_2, ..., c_m, all not zeros, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_m\mathbf{v}_m = \mathbf{0}.$$

Chapter 5: More About Eigenvalues and Eigenvectors

E1. Definition and properties

T is said to be a nilpotent operator if $T^m = 0$ for some non-negative integer m.

Examples:

- $T(a, b, c, d) = (0, a, b, c)$
- Differentiation on $P_5(\mathbb{R})$

E1. Definition and properties

It is easy to see that

If T is nilpotent, then

(a) $T^n = 0$;

(b) Every non-zero vector $\mathbf{v} \in V$ is a 0-generalized eigenvalue of T;

(c) 0 is the only eigenvalue of T.

Is the converse true?

E1. Definition and properties

The following are equivalent.

(a) T is nilpotent.

(b) The minimal polynomial of T is t^m for some positive integer m.

(c) There exists a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is upper triangular with zeros on the diagonal.

if $\exists \sum_{i=0}^{n<m} x^i t^i = p = 0$

then $t^{m-n} p = 0$. and $t^{m-n} \cdot t^n = 0$

i.e. $t^{m-n} \sum_{i=0}^{n-1} x^i t^i = 0$

...

$t = 0$.

E2. Generalised eigenspaces

If $\mathbf{v}$ is a generalised eigenvector of T corresponding to λ, we define the generalised eigenspace of T corresponding to λ, denoted $G(\lambda, T)$, to be

$$G(\lambda, T) = \{\mathbf{v} \in V : (T - \lambda I)^k(\mathbf{v}) = \mathbf{0} \text{ for some positive integer } k\}.$$

Note that $G(\lambda, T)$ is simply the kernel of $\underline{(T-\lambda I)^{\dim V}}$.

Recall the example $T(x, y, z) = (4y, 0, 5z)$.

E2. Generalised eigenspaces

Suppose $\mathbb{F} = \mathbb{C}$ and λ_1, λ_2, ..., λ_m are the distinct eigenvalues of T. Then

(a) $G(\lambda_k, T)$ is T-invariant for every k;

(b) $(T - \lambda_k I)_{G(\lambda_k, T)}$ is nilpotent for every k;

(c) $V = G(\lambda_1, T) \oplus G(\lambda_2, T) \oplus \cdots \oplus G(\lambda_m, T)$.

(a) $\exists\, v \in G(\lambda_k, T)$

$T(T - \lambda_k I)^m v = 0 = (T^m - \lambda_k I)(Tv)$.

$Tv \in \text{null}\,(T^m - \lambda_k I)^m$

$\dim V$

(b) $\left((T - \lambda_k I)|_{G(\lambda_k, T)}\right)^{\dim V} = 0$.

(c)

E3. Block diagonal form

Although not every operator T is diagonalisable, the above results imply that we can find a basis $\mathcal{B}$ of V for which $[T]_{\mathcal{B}}$ is in block diagonal form where each diagonal block is square and upper triangular.

Illustrate with the example

$$\begin{bmatrix} 6 & 0 & 0 \\ 3 & 6 & 0 \\ 4 & 2 & 7 \end{bmatrix}$$

$$T(x, y, z) = (6x + 3y + 4z, 6y + 2z, 7z)$$

from Chapter 3, where we found that the eigenvalues of T are 6 and 7.

E4. More about square roots of operators

If T is nilpotent, then $I + T$ has a square root.

Proof:

Suppose $T^m = 0$. We look for a square root of T of the form

$$a_0 I + a_1 T + a_2 T^2 + \cdots + a_{m-1} T^{m-1}.$$

$\sim^2 = a_0^2 I + 2a_0a_1 T + (2a_0a_2 + a_1^2) T^2 + \cdots (2a_{m-1}a_0 + 2a_{m-2}a_1 + \cdots +) T^{m-1}$

let $a_0 = 1$

$a_1 = \frac{1}{2}$

$a_2 = \frac{-a_1^2}{2a_0} = \frac{1}{8}$

$\cdots$

E4. More about square roots of operators

If $\mathbb{F} = \mathbb{C}$, every invertible operator T has a square root.

Proof:

For each eigenvalue λ_k of T, we have

$$T_{G(\lambda_k,T)} = \lambda_k I + T_k \qquad \text{where } T_k \text{ is nilpotent.}$$

Such operator thus has a square root.

Chapter 5: More About Eigenvalues and Eigenvectors

F1. Algebraic and geometric multiplicities

Let λ be an eigenvalue of T. We define the following:

$$\text{Algebraic multiplicity of } \lambda = \dim G(\lambda, T)$$

$$\text{Geometric multiplicity of } \lambda = \dim E(\lambda, T)$$

Remarks:

- Some authors use the term 'multiplicity' for 'algebraic multiplicity'.
- It is clear from the definition that the algebraic multiplicity is always bigger than or equal to the geometric multiplicity.
- IF $\mathbb{F} = \mathbb{C}$, the sum of the algebraic multiplicities of all eigenvalues of T must be equal to dim V.

F1. Algebraic and geometric multiplicities

Consider again the example

$$\begin{bmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{bmatrix}$$

$$T(x, y, z) = (6x + 3y + 4z, 6y + 2z, 7z)$$

where we knew that the eigenvalues of T are 6 and 7. What are the algebraic and geometric multiplicities of the eigenvalues?

F1. Algebraic and geometric multiplicities

Suppose $\mathbb{F} = \mathbb{C}$ and let $\mathcal{B}$ be a basis of V for which $[T]_{\mathcal{B}}$ is upper triangular. Then the diagonal entries of $[T]_{\mathcal{B}}$ are the eigenvalues of T appearing with the same (algebraic) multiplicities.

Proof:

It suffices to show that, if λ is an eigenvalue of T which occurs d times on the diagonal of $[T]_{\mathcal{B}}$, then $\ker(T - \lambda I)^n$ has dimension at most d.

Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and λ_i be the i-th diagonal entry of $[T]_{\mathcal{B}}$. For each k,

$$T(\mathbf{v}_k) = \mathbf{u}_k + \lambda_k \mathbf{v}_k \qquad \text{for some } \mathbf{u}_k \in \operatorname{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}\}.$$

Suppose 0 occurs d times on the diagonal of $[T]_{\mathcal{B}}$. Then nullity $T \underline{\;\leq\;}$ d.

F2. Characteristic polynomial

Suppose λ_1, λ_2, ..., λ_m are the distinct eigenvalues of T, with algebraic multiplicities d_1, d_2, ..., d_m respectively. We define the characteristic polynomial of T, denoted χ_T, by

$$\chi_T(t) = (t - \lambda_1)^{d_1}(t - \lambda_2)^{d_2} \cdots (t - \lambda_m)^{d_m}.$$

Clearly $\chi_T(t)$ is monic and of degree n, and that its zeros are precisely the eigenvalues of T with the same (algebraic) multiplicities.

F2. Characteristic polynomial

Cayley-Hamilton Theorem

When $\mathbb{F} = \mathbb{C}$, $\chi_T(T) = 0$ (hence $m_T \mid \chi_T$).

Proof:

Suppose

$$\chi_T(t) = (t - \lambda_1)^{d_1}(t - \lambda_2)^{d_2} \cdots (t - \lambda_m)^{d_m}.$$

It suffices to show that for any k, $\chi_T(T)(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in G(\lambda_k, T)$.

$\dim G(\lambda_k, T) = d_k$ and $(T - \lambda_k)|_{G(\lambda_k, T)}$ is nilpotent.

i.e. $\left((T-\lambda_k)_{G(\lambda_k,T)}\right)^{d_k} = 0$.

F3. Jordan basis

Show that the following operators are not diagonalisable.

(a) $T : \mathbb{F}^4 \to \mathbb{F}^4$ defined by

$$T(a, b, c, d) = (0, a, b, c)$$

(b) $T : \mathbb{F}^6 \to \mathbb{F}^6$ defined by

$$T(a, b, c, d, e, f) = (0, a, b, 0, d, 0)$$

F3. Jordan basis

Despite T being non-diagonalisable, can you find a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}$ is nice?

(a) $T : \mathbb{F}^4 \to \mathbb{F}^4$ defined by

$$T(a, b, c, d) = (0, a, b, c)$$

(b) $T : \mathbb{F}^6 \to \mathbb{F}^6$ defined by

$$T(a, b, c, d, e, f) = (0, a, b, 0, d, 0)$$

F3. Jordan basis

$$A \to \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & 1 & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}$$

A basis $\mathcal{B}$ of V is said to be a Jordan basis for T if

$$[T]_{\mathcal{B}} = \begin{bmatrix} A_1 & O & \cdots & O \\ O & A_2 & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_k \end{bmatrix},$$

where each A_i is a square matrix such that

- all entries on the main diagonal are the same;
- all entries that are immediately above the main diagonal are 1; and
- all other entries are 0.

F3. Jordan basis

We shall prove that

> Every nilpotent operator T has a Jordan basis.

Note that this implies

> When $\mathbb{F} = \mathbb{C}$, every operator T has a Jordan basis.

F3. Jordan basis

We prove by induction on n that every nilpotent operator T has a Jordan basis. The result holds for $n = 1$. Assume $n > 1$ and the result holds for smaller n.

Let m be the smallest positive integer such that $T^m = 0$. Pick $\mathbf{u} \in V$ such that $T^{m-1}(\mathbf{u}) \neq \mathbf{0}$. Let U be the span of the linearly independent vectors

$$\mathbf{u}, \ T(\mathbf{u}), \ T^2(\mathbf{u}), \ \ldots, \ T^{m-1}(\mathbf{u}).$$

If $U = V$, we are done. Otherwise, note that U is T-invariant, so it suffices to find a T-invariant subspace W of V such that $V = U \oplus W$.

F3. Jordan basis

Pick $f \in V'$ such that $f(T^{m-1}(\mathbf{u})) \neq 0$. Define

$$W = \{\mathbf{v} \in V : \mathbf{v}, T(\mathbf{v}), ..., T^{m-1}(\mathbf{v}) \in \ker f\}.$$

We check that

- W is a T-invariant subspace of V;
- $U \cap W = \{\mathbf{0}\}$; and
- $\dim U + \dim W = n$.